CAUSAL DISSECTION OF SCIENTIFIC AGENTS: BREAKS FROM EVIDENCE TO ACTION

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ABSTRACT

Experimental success does not reveal whether a scientific agent learned the right model. The same endpoint can follow from an inherited prior, productive search, phenomenological interpolation, or evidence-driven revision. We use ChemWorld as a controlled causal probe: the executable world, interface, resources, and stochastic identity remain fixed while the agent's initial model is made opaque, aligned, or misspecified at entity, parametric, or structural loci. Across 135 public agent sessions, these interventions redirected search and prediction errors fell in every arm and locus, yet the prespecified selective-correction criterion failed at all three loci. Under matched structural counterevidence, mean normalized errors in all three arms fell from 0.2255-0.3392 to 0.0060-0.0074, but 0/5 misspecified summaries recovered the registered 1.75 power law. All 135 typed laws executed, yet 84 preserved less predictive information than the agents' final explicit predictions. In fresh multi-task action selection, only 11/42 eligible readouts selected the best unseen plan, and thresholded law adequacy did not determine a correct action. Finally, a complete-ranking control passed 7/8 fresh 96-query units while selecting Top-1 in only 1/8, whereas a fresh 320-query unit failed the same gate but selected Top-1 with zero regret. Experimental optimization, numerical revision, structural identification, executable compression, decision transfer, and evaluator validity are therefore distinct transitions rather than one score.

1 INTRODUCTION

Autonomous experimental agents are increasingly judged by whether they discover high-performing molecules or reactions. Yet an endpoint benchmark cannot distinguish whether an agent inherited a fortunate prior, navigated a productive search corridor, adopted a phenomenological interpolation, or genuinely revised its scientific understanding from evidence. We call this ambiguity endpoint underdetermination. The relevant question is not only whether an agent finds a good experiment, but what that experiment changes in its model of the world.

This distinction matters as language-model agents plan syntheses, call chemistry tools, operate instruments, and enter self-driving laboratory workflows (Boiko et al., 2023; Bran et al., 2024; Szymanski et al., 2023; Darvish et al., 2025; Song et al., 2025; Vriza et al., 2026). Interactive environments increasingly test repeated cycles of hypothesis, intervention, and inference (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Zheng et al., 2026; Yang et al., 2026; Batzoglou, 2026). Yet pretrained knowledge, prompt-provided information, experiment selection, endpoint optimization, and verbal explanation are usually observed together. Consequently, a high score need not identify which transformation succeeded, and a low score need not identify which transformation failed.

We make those transformations observable by using ChemWorld as a controlled causal probe. The same executable world, interface, resources, and stochastic identity are held fixed while its initial model is made opaque, aligned, or misspecified at a declared entity, parametric, or structural locus. Persistent agents then interact through a transactional laboratory interface. Evaluator-owned counterfactual queries, matched contradictory evidence, executable-law assays, and unseen complete ActionPlans separately measure search, prediction, structural identification, compression, and decision transfer.

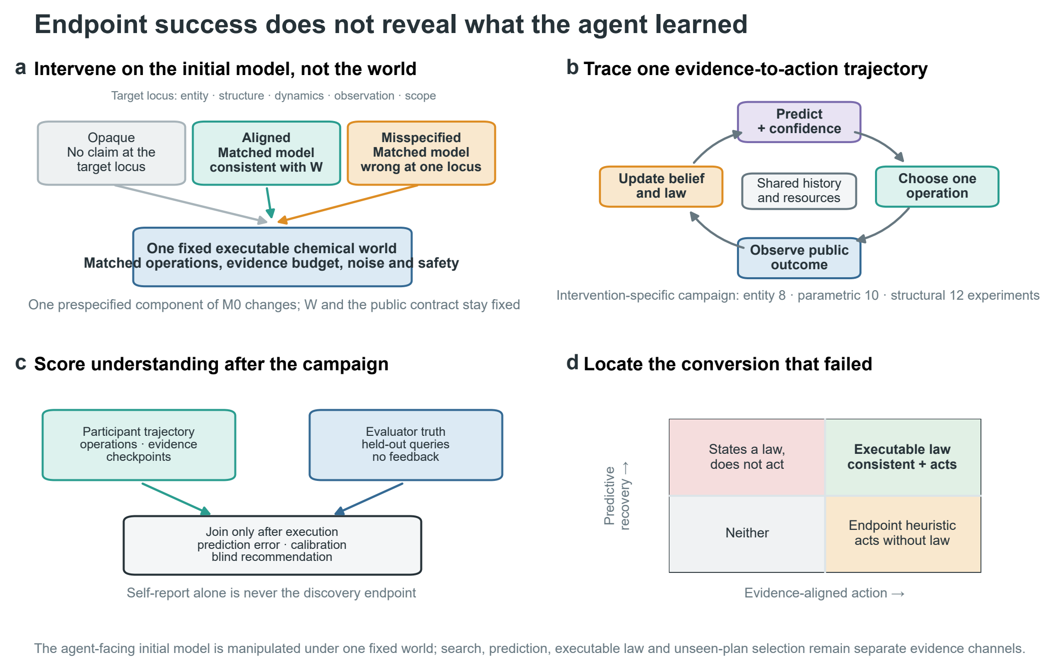

Figure 1: **Endpoint success underdetermines scientific competence.** The same outcome can arise from a correct prior, local search, or evidence-driven model correction. ChemWorld intervenes on the initial model while holding the executable world fixed, then evaluates the transitions from evidence to predictions, executable laws, and unseen actions.

Our contribution is a causal map of the evidence-to-action chain rather than another scalar leaderboard. We identify four breaks. Priors redirect trajectories, but the registered selective-correction criterion fails at every locus. Matched counterevidence drives numerical predictions near truth, yet 0/5 misspecified summaries recover the registered power law. Executable laws run successfully but are less faithful than final explicit predictions in 84/135 sessions. The fourth break has two levels: agent-level law adequacy does not determine unseen Top-1 selection, and evaluator-level complete-ranking qualification does not determine action validity. The resulting map locates losses in revision, structural identification, compression, decision transfer, and the evaluator itself without treating every failure as low overall capability.

2 Environment and Evaluation

2.1 ChemWorld as a controlled causal probe

ChemWorld separates three objects. The executable world contains hidden state, transition and measurement laws, resources, and terminal assays. The public task exposes typed operations, instruments, observations, and an initial model. The evaluator owns held-out truth, replay, and scoring. For world w and initial-model arm a, the hidden transition process remains fixed while the supplied model $M_{0,a}$ changes. The aligned model is correct at one declared locus, the misspecified model is explicit but wrong at that locus, and the opaque arm withholds the corresponding structure. Because operations and measurements remain identical, differences can be attributed to information supplied to the complete agent–tool system rather than to different physics.

Actions are transactions, not free-form prose. The host validates a typed request, checks resources and preconditions, commits the state transition, and returns a public observation or a structured rejection. Failed actions and discards remain in the trajectory. Every submitted action, commit, rollback, measurement, assay, and resource delta is recorded for exact replay. These semantics matter: an action recommendation is scientific evidence only when the evaluator can execute the same complete plan without filling in hidden workflow choices.

Table 1: **Executed evidence layers and claim boundaries.** A completed work package may contain a retained scientific rejection; unstarted units are not silently removed.

Layer	Units	Execution	Supported role
Prospective priors	135 sessions	1,243/1,260 experiments	Search, prediction, law, incumbent replay
Matched evidence	30 sessions	fixed evidence packets	Numerical update versus structural recovery
Open action	45 cells	42 eligible; 240/240 truth/replay	Descriptive unseen-plan selection
96-query control	15 planned	8 attempted; 896/896 truth/replay	Retained pre-participant rejection
320-query control	7 exposed + 15 fresh	7 exposed pass; 1 fresh attempted	Construction repair versus prospective rejection
Gate alignment	16 unit versions	0 new execution	Rank validity versus action validity

2.2 Five-stage capability chain

We distinguish five downstream outcomes. **Evidence acquisition** asks which informative experiments are selected and observed. **Numerical revision** asks whether held-out counterfactual prediction error falls. **Structural identification** asks whether the agent rejects a false form and recovers the registered relation. **Executable compression** asks whether a typed law preserves the information in conditional predictions. **Decision transfer** asks whether the final state selects a good previously unseen plan. A sixth requirement, **evaluator validity**, asks whether the control and metric actually measure the decision estimand. Success at one stage is neither defined as nor assumed to imply success at the next.

At five checkpoints per session, the participant reports initial-model reliability, predictions, uncertainty, evidence references, an executable law, and the next experimental intent. The evaluator executes prespecified counterfactual query sets independently of the participant. It scores 675/675 checkpoints from 420/420 truth executions and later runs each final typed law on the same coordinates. Paired blind replay evaluates the final recommendation against the observed incumbent, while a separate longitudinal assay reveals eight outcome-hidden complete ActionPlans only after autonomous exploration has ended.

3 Experimental programme

The prospective programme is layer-stratified. Entity interventions cover five task families and five independently selected public worlds per task. Parametric and structural blocks each cover two validated task families and five worlds. Every task–world cluster contains opaque, aligned, and misspecified arms, yielding 45 clusters and 135 independent persistent sessions. Campaign length is locus-specific: eight, ten, or twelve complete experiments, with five checkpoints in every session. Exploratory, validation, prospective, matched-evidence, and open-action worlds remain separated.

The primary contrast tests selective evidence-driven correction. If $E^{(\ell)}_{a,k}$ is held-out error for arm a, checkpoint k, and locus ℓ, then

$$C_\ell = \left(E^{(\ell)}_{\mathrm{mis},0} - E^{(\ell)}_{\mathrm{mis},K}\right) - \left(E^{(\ell)}_{\mathrm{aligned},0} - E^{(\ell)}_{\mathrm{aligned},K}\right).$$

Success requires greater correction in the misspecified arm, improvement of that arm, and no material deterioration of the aligned arm. Loci are decided separately; unlike intervention semantics are not pooled. Failed scientific cells stay in their scheduled denominator, with the last valid checkpoint carried forward for right-censoring and zero primary improvement assigned when the final prediction is missing. Only a pure infrastructure failure without a persisted trajectory can resume under a fixed attempt cap.

Matched-evidence sessions use cloned worlds and provide the same decisive counterevidence to every arm, separating failure to seek evidence from failure to update after seeing it. The longitudinal action matrix separately contains three tasks, five worlds, and three arms (45 scheduled cells). After 12 autonomous experiments, each agent ranks eight new plans; regret and Top-1 are primary action readouts, while complete-rank correlation and law adequacy are diagnostics. Table~1 keeps these layers and their claim boundaries explicit.

4 RESULTS: FROM PRIORS TO LAWS

4.1 PRIORS SCAFFOLD SEARCH WITHOUT SELECTIVE CORRECTION

All 135 scheduled sessions produced final records. Participants completed 1,243/1,260 planned experiments; 121 sessions met operational eligibility. The denominator retains 26 discarded lifecycles, 13 resource-ledger rejections, and all right-censored cells. Every session submitted five checkpoints, providing 6,300 counterfactual predictions and 24,300 query–metric values.

The intervention reached behavior. The first complete recipe differed between aligned and misspecified cells in 45/45 matched clusters, between opaque and aligned in 45/45, and between opaque and misspecified in 44/45. This is a manipulation check rather than a causal effect estimate because there are no repeated same-arm sessions. Search continued after the first proposal: 91.2% of completed experiments used a unique recipe, 84.4% of session optima appeared after the midpoint, and 32.6% appeared in the last completed experiment.

Correct-prior utility was task dependent. In entity-level partition, aligned information produced a +0.200 best-endpoint advantage over the misspecified arm in 5/5 worlds. In structural crystallization, a +0.141 first-experiment head start narrowed to +0.055 as the disadvantaged arm explored. In structural partition, aligned and misspecified descriptions both helped relative to opaque identifiers while differing little from one another. A correct model therefore altered the search landscape without imposing one endpoint ordering.

Prediction error nevertheless fell on average in every arm at every locus. Reductions for opaque, aligned, and misspecified cells were 0.111/0.097/0.097 at the entity locus, 0.090/0.033/0.065 at the parametric locus, and 0.219/0.228/0.221 at the structural locus. The stricter selective-correction contrasts were -0.214 for entity ($p = 0.990$), +0.033 for parametric ($p = 0.079$), and -0.224 for structural ($p = 1.000$); none passed. General predictive learning did not become preferential correction of a wrong starting model (Fig.~2).

4.2 MATCHED COUNTEREVIDENCE SUPPORTS INTERPOLATION WITHOUT STRUCTURAL RECOVERY

Free discovery cannot distinguish failure to seek diagnostic evidence from failure to use it. In the parametric matched-evidence block, all five misspecified summaries rejected the supplied high-potential direction and recovered the peak-and-collapse response once the decisive profile was provided. Part of the free-discovery loss therefore lay in evidence acquisition.

The corrected structural assay showed a different bottleneck. After every arm received the same direct phase-process evidence, mean normalized errors fell from 0.2255, 0.2736, and 0.3392 to 0.0074, 0.0060, and 0.0071 for opaque, aligned, and misspecified cells. The misspecified-minus-aligned update contrast was positive in only 3/5 worlds (exact one-sided sign-flip $p = 0.125$), so this is numerical convergence rather than a confirmatory arm effect. Yet 0/5 misspecified public summaries recovered the registered 1.75 power law, 1/5 explicitly rejected the supplied linear form, and 5/5 adopted a saturation or endpoint model. Direct evidence supported accurate local phenomenological interpolation without structural identification (Fig.~3).

All 135 final typed laws executed on their prespecified coordinates, but executability did not preserve the information in conditional predictions. Mean law error was 0.237; relative to effective final predictions, laws were better in 50 cells, equal in one, and worse in 84, with mean law-minus-prediction error +0.069. Typed syntax solved execution coverage, not faithful scientific compression. A same-domain capacity oracle then fitted legal typed laws directly to each complete final prediction vector. The full registered basis reproduced 135/135 prediction states (mean MAE 4.25×10^{-13}), compared with 0.1539 MAE for participant laws; participant-term-matched and leave-one-query-out fits reached 0.0114 and 0.0788 MAE. All fitted laws passed the production parser/executor. This localizes most loss to agent distillation rather than interface capacity, while remaining an in-domain capacity control rather than global mechanism recovery. Paired blind replay likewise tested only incumbent retrieval: 726/726 executions completed for 121 evaluable cells, with recommendations better/equivalent/worse in 1/119/1 cells. Selection beyond the observed campaign required a separate assay.

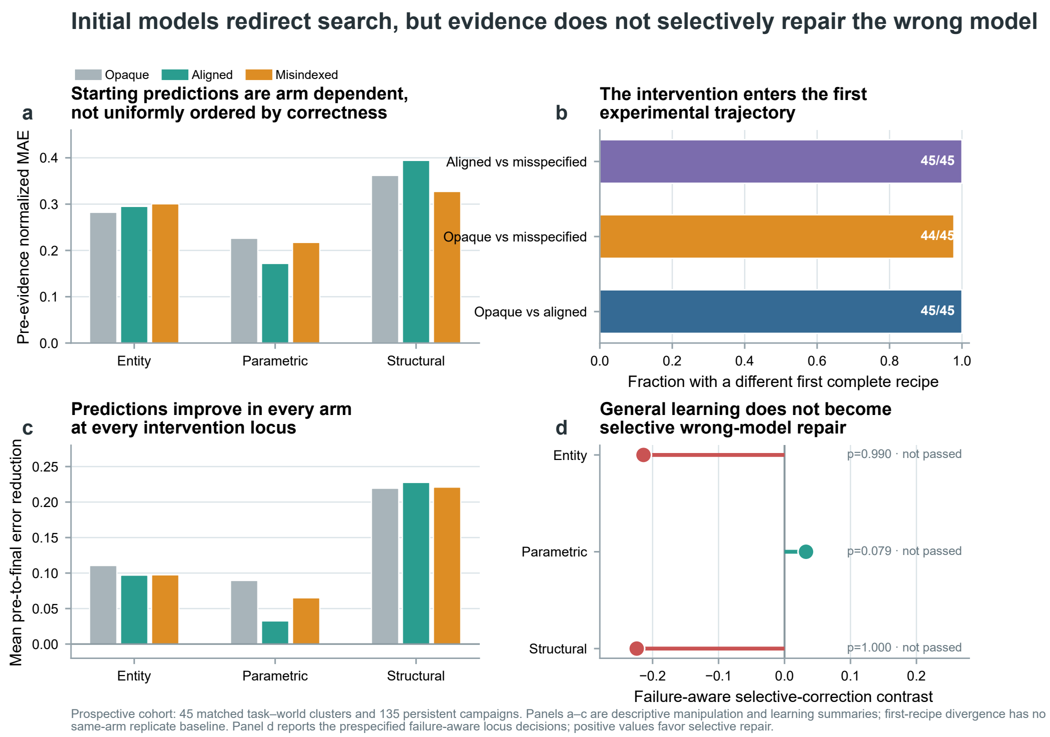

Figure 2: **Initial models redirect search, but evidence does not selectively repair the wrong model.** Pre-evidence errors, first-recipe divergence, pre-to-final error reductions, and frozen failure-aware selective-correction contrasts are reported by intervention locus. Every arm improves on average, but no locus passes the selective-correction criterion.

5 RESULTS: LAW, ACTION, AND EVALUATION

5.1 AGENT-LEVEL LAW ADEQUACY DOES NOT DETERMINE UNSEEN ACTION SELECTION

The longitudinal matrix produced records for 45/45 scheduled cells. Truth evaluation and exact replay both completed 240/240 plan executions. Forty-two cells were uncontaminated and eligible for action metrics; two agent-induced resource/process failures and one session interruption in crystallization remain in the scheduled denominator.

Only 11/42 eligible terminal readouts selected the true Top-1 plan. Mean selected rank was 3.31 of 8 and mean normalized regret was 0.297. The uniform-random rank 4.5 is geometric context, not a no-evidence control, so these outcomes describe post-campaign competence rather than a causal benefit of exploration. Law and action also separated: 30 cells had an inadequate law and wrong action, 11 an inadequate law but correct action, one an adequate law but wrong action, and none an adequate law and correct action. Task means ranged from rank 2.00 for reaction safety to 4.58 for crystallization, and every pairwise arm contrast changed sign under some leave-one-cluster-out omission. We therefore make no pooled arm-level claim (Fig.~4a,b).

This separation survives continuous and threshold-sensitive analysis. Pooled law MAE correlated weakly with selected rank (Spearman $\rho = -0.073$, cluster-bootstrap 95% interval $[-0.380, 0.256]$) and normalized regret ($\rho = -0.133$, $[-0.452, 0.217]$), while task-specific rank associations reversed sign ($+0.524$, -0.592, and -0.007). Across law-MAE thresholds from 0.05 to 0.30, the adequate subset expanded from 1 to 34 cells but contained only 0 to 9 correct actions. The four-way table is therefore not a single-cutoff artifact: law error has no stable monotonic map to unseen-action quality across tasks.

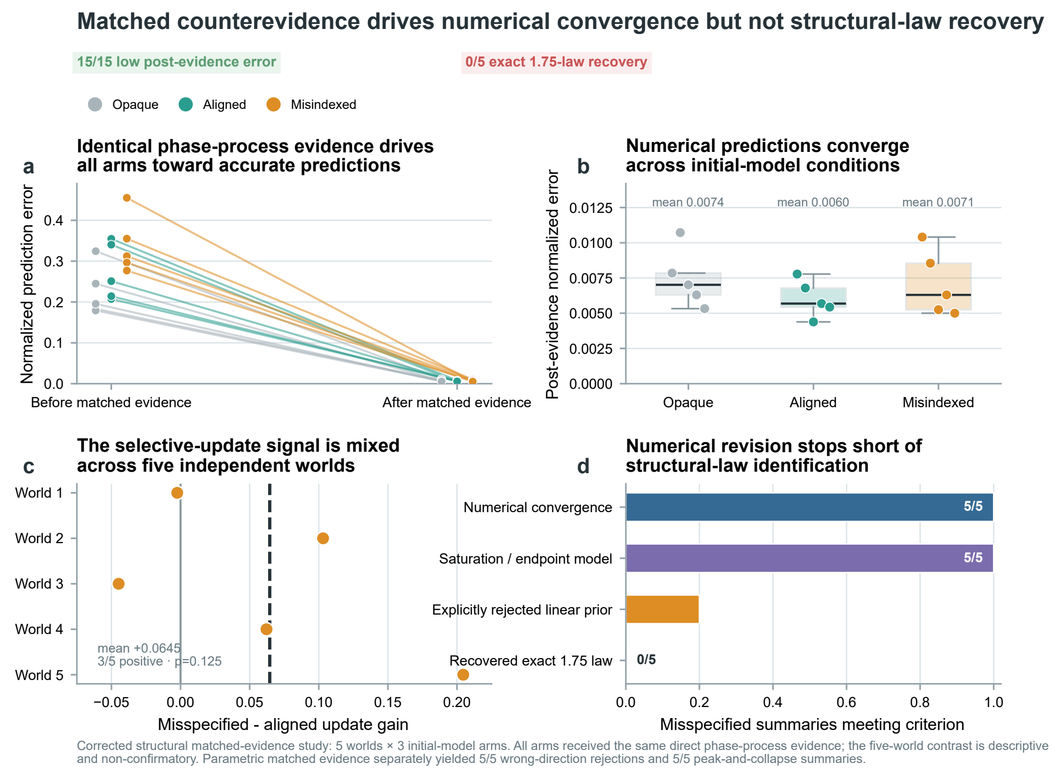

Figure 3: **Matched counterevidence drives numerical convergence but not structural-law recovery.** All 15 corrected structural cells receive identical evidence and converge near truth. Across the five misspecified worlds, exact recovery of the registered power law is absent despite low post-evidence error.

5.2 Evaluator-level ranking and decision quality are different estimands

A planned five-condition causal follow-up required a provider-free oracle law to rank the eight candidates with Spearman $\rho \geq 0.80$ before any participant was called. The first eight of 15 fresh clusters completed 896/896 truth executions and exact replays. Candidate-opportunity gates passed 8/8, but the eighth oracle obtained $\rho = 0.738095$ and disagreed on Top-1. The frozen stop rule left seven clusters, the operational canary, all 225 participant sessions, and 540 planned participant experiments unstarted. This is a rejection of the control construction, not an estimated participant effect.

We then isolated grid coverage while retaining the fitted ExtraTrees family and the correlation threshold. Expanding from 96 queries to 64 global plus 256 candidate-neighborhood queries repaired all four historical failures: 7/7 exposed construction units passed with 2,352/2,352 truth executions and exact replays, and minimum ρ was 0.857143. Prospective validity did not follow. The first new electrochemical world completed 336/336 truth executions and replays but obtained $\rho = 0.714286$, triggering the same stop boundary and leaving 14 fresh clusters unstarted. Nevertheless, this oracle selected the true Top-1 action with normalized regret zero.

A frozen zero-execution diagnostic reproduced the original rank and Top-1 outcomes for all 16 completed 96- and 320-query unit versions. Among the eight fresh 96-query units, the rank gate passed 7/8 but Top-1 was correct in only 1/8 and within 0.01 of optimum in 3/8; six were rank-pass/action-wrong. The fresh 320-query unit was rank-fail/action-correct. Thus global-order correlation and decision validity disagree bidirectionally. The historical gates and stops remain valid for their protocols, but future controls must prospectively prioritize regret, near-optimal selection, and near-tie-aware ordering, retaining full-ranking correlation as a secondary diagnostic.

Table 2: **Complete-ranking gates and action endpoints are different estimands.**

Frozen block	n	Rank pass	Top-1	≤ 0.01 regret
Fresh 96-query	8	7	1	3
Exposed 320-query	7	7	4	6
Fresh 320-query	1	0	1	1

Figure 4: **Agent law-action decoupling and evaluator rank-action misalignment. a,** True rank for 42 eligible W2-50 cells; all 45 scheduled cells remain in the denominator. **b,** Continuous law error versus regret, with Top-1 stars and a seven-cutoff inset. **c,** Frozen W2-51/W2-52 qualification dispositions separate exposed construction from fresh qualification; participant execution is zero. **d,** Spearman correlation versus regret for all 16 frozen W2-53 unit versions. No panel revises a historical stop decision.

For eight candidates without ties, Spearman correlation is $\rho = 1 - \sum_i d_i^2/84$, where d_i is the displacement of candidate i between predicted and true ranks. The retained 96-query failure, $\rho = 0.738095$, therefore represents $\sum_i d_i^2 = 22$; the fresh 320-query failure, $\rho = 0.714286$, represents 24. The scalar says how much the full ordering moved, not where the displacement occurred. The former selected the wrong Top-1 while the latter preserved it exactly, making the estimand mismatch concrete rather than merely threshold dependent (Fig.~4c,d; Table~2).

6 Related work

Self-driving laboratories and chemistry agents emphasize closed-loop execution, tool use, or endpoint optimization (Felton et al., 2021; Häse et al., 2021; Boiko et al., 2023; Bran et al., 2024). Virtual laboratories and process-control environments provide scalable interaction and safety (Beeler et al., 2024; Bloor et al., 2024; Malik et al., 2026; Chen et al., 2025). Scientific-discovery benchmarks increasingly test iterative experimentation, causal inference, and transferable knowledge (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Yang et al., 2026; Batzoglou, 2026). ChemWorld complements these directions by making the initial model an experimental variable and by evaluating each transformation from evidence to action with evaluator-owned truth and exact replay. The central distinction is diagnostic: endpoint success, prediction learning, law recovery, and action quality are not interchangeable readouts.

7 Discussion and limitations

The evaluated system frequently searches productively and learns numerically, but its transformations lose information at different locations. Correct priors alter trajectories without a general endpoint advantage. Free evidence reduces prediction error without selectively repairing wrong priors. Direct matched evidence can yield near-exact numerical revision while all five misspecified summaries remain phenomenological saturation or endpoint models instead of recovering the registered power law. Typed laws can execute yet compress predictions poorly. At the agent level, law adequacy does not determine the correct unseen action; at the evaluator level, complete-ranking qualification does not determine action validity. These are distinct capabilities with different interventions and denominators.

The oracle study extends the second level of this diagnosis to the harness. A metric can be reproducible and still be misaligned with the intended decision. Spearman correlation rewards ordering across all candidates, whereas the causal follow-up needs a positive control for selected action and decision loss. Stopping before participant execution prevented a weak control from producing an uninterpretable effect. Evaluator validity is part of scientific-agent validity, not a post-hoc implementation detail.

7.1 A transition map for scientific-agent evaluation

The capability chain suggests three design requirements. First, intervention and readout need separate units: manipulating priors, supplying matched evidence, and hiding candidate outcomes answer different questions. Prediction, law execution, and action regret should remain separate so failures can be localized to acquisition, revision, representation, or transfer.

Second, controls should qualify on the intended endpoint. An outcome-disjoint, exactly replayable law can rank globally well yet choose the wrong action; an optimal decision can coexist with moderate ranking error. Selection studies should register regret, near-optimality, and near-tie handling as primary, leaving full-rank metrics diagnostic.

Third, evidence roles and denominators must remain separate. The 96-query failure tests a control; the 320-query exposed pass tests construction; the fresh 320-query failure tests generalization. Pooling them or treating stopped sessions as zero effects would invent evidence; scheduled, eligible, completed, and unstarted counts prevent this error.

7.2 Limitations and conclusion

Evidence concerns one agent–tool system in simulation, not cross-model ranking or laboratory fidelity; some contrasts use five worlds. The open-action matrix lacks no-evidence and pre-exploration baselines, and the five-condition cohort has no participant data. Artifact portability, private replication, and cross-provider generalization remain untested.

Scientific-agent claims should name interventions, units, failure rules, and transfer boundaries. Agency is a chain from evidence to revision, identification, compression, and decision, breakable in the agent or its evaluator.

AI use statement

Generative AI tools assisted software implementation, experiment orchestration, data analysis, visualization code, literature discovery, and drafting and editing of the manuscript. The authors reviewed the study designs and frozen decision rules, inspected retained failures, verified reported denominators and claims against machine-readable records, and tested AI-assisted code before use. The authors take responsibility for the final content, including text, claims, code, and artifacts produced with generative-AI assistance.

Reproducibility statement

The main paper defines the experimental units, interventions, estimands, failure handling, and stop boundaries. The appendix provides the execution and evaluation details needed to reconstruct each evidence layer. Exact action traces, evaluator-owned truth queries, replay checks, complete failure records, and machine-readable summaries will be included in the anonymous supplementary package.

Ethics statement

The experiments use simulated chemical environments and no human participants, personal data, or live laboratory actuation. The principal risks are over-interpreting simulated results and treating an evaluator surrogate as the scientific objective; the claim boundaries and retained negative qualification results are reported to mitigate these risks.

References

Serafim Batzoglou. ReplaySCM: A benchmark for executable causal mechanism induction from interventions, 2026. Preprint.

Chris Beeler, Sriram Ganapathi Subramanian, Kyle Sprague, et al. ChemGymRL: A customizable interactive framework for reinforcement learning for digital chemistry. *Digital Discovery*, 3: 742–758, 2024. doi: 10.1039/D3DD00183K.

Maximilian Bloor, José Torraca, Ilya Orson Sandoval, et al. PC-Gym: Benchmark environments for process control problems, 2024. Preprint, arXiv:2410.22093.

Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. doi: 10.1038/s41586-023-06792-0.

Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6:525–535, 2024. doi: 10.1038/s42256-024-00832-8.

Yimeng Chen, Piotr Piękos, Mateusz Ostaszewski, Firas Laakom, and Jürgen Schmidhuber. PhysGym: Benchmarking llms in interactive physics discovery with controlled priors. In *Advances in Neural Information Processing Systems*, 2025. URL `https://openreview.net/forum?id=w8uII2qAmd`.

Kourosh Darvish, Marta Skreta, Yuchi Zhao, et al. ORGANA: A robotic assistant for automated chemistry experimentation and characterization. *Matter*, 8(2), 2025. doi: 10.1016/j.matt.2024.10.015.

Haonan Duan, Stephen Zhewen Lu, Caitlin Fiona Harrigan, Nishkrit Desai, Jiarui Lu, Michał Koziarski, Leonardo Cotta, and Chris J. Maddison. Measuring scientific capabilities of language models with a systems biology dry lab, 2025. Preprint, arXiv:2507.02083.

Kobi C. Felton, Jan G. Rittig, and Alexei A. Lapkin. Summit: Benchmarking machine learning methods for reaction optimisation. *Chemistry–Methods*, 1:116–122, 2021. doi: 10.1002/cmtd.202000051.

Kanishk Gandhi, Michael Y. Li, Lyle Goodyear, Agam Bhatia, Louise Li, Aditi Bhaskar, Mohammed Zaman, and Noah D. Goodman. BoxingGym: Benchmarking progress in automated experimental design and model discovery, 2025. Preprint, arXiv:2501.01540.

Florian Häse, Matteo Aldeghi, Riley J. Hickman, Loïc M. Roch, Melodie Christensen, Elena Liles, Jason E. Hein, and Alán Aspuru-Guzik. Olympus: a benchmarking framework for noisy optimization and experiment planning. *Machine Learning: Science and Technology*, 2:035021, 2021. doi: 10.1088/2632-2153/abedc8.

Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Oyvind Tafjord, and Peter Clark. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0324.

Shreshth A. Malik, Tiarnan Doherty, Panagiotis Tigas, Muhammed Razzak, Stephen J. Roberts, Aron Walsh, and Yarin Gal. MADE: Benchmark environments for closed-loop materials discovery. In *International Conference on Machine Learning*, 2026.

Tao Song, Man Luo, Xiaolong Zhang, et al. A multiagent-driven robotic AI chemist enabling autonomous chemical research on demand. *Journal of the American Chemical Society*, 147(15): 12534–12545, 2025. doi: 10.1021/jacs.4c17738.

Nathan J. Szymanski et al. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature*, 624:86–91, 2023. doi: 10.1038/s41586-023-06734-w.

Aikaterini Vriza, Michael H. Prince, Tao Zhou, Henry Chan, and Mathew J. Cherukara. Operating advanced scientific instruments with AI agents that learn on the job. *npj Computational Materials*, 12:160, 2026. doi: 10.1038/s41524-026-02005-0.

Junlin Yang, Dylan Zhang, Xiangchen Song, et al. CausaLab: A scalable environment for interactive causal discovery toward ai scientists, 2026. Preprint.

Tianshi Zheng, Kelvin Kiu-Wai Tam, Newt Hue-Nam K. Nguyen, et al. NewtonBench: Benchmarking generalizable scientific law discovery in llm agents. In *International Conference on Learning Representations*, 2026.

A Appendix

A.1 Environment semantics and intervention construction

An executable world is a stateful transition system with typed operations, measurement functions, finite resources, terminal assays, and an evaluator-only truth surface. A task publishes the operation and instrument contracts while withholding the latent state and registered relations. A scenario binds a world, public task description, resource card, and initial model. Initial-model arms alter one declared information locus without changing the transition process, measurement process, candidate actions, or evaluator truth.

The entity locus changes material identity or ontology. The parametric locus changes a quantitative response model while preserving the declared entities. The structural locus changes the functional form used to describe a mechanism. Aligned and misspecified models are matched in explicitness so that a contrast is not merely information versus no information. The opaque arm exposes the same laboratory interface but withholds the intervened model component. Exploratory, validation, prospective, and private world selections are disjoint by construction.

A.2 Persistent execution and checkpoints

Each participant cell is one persistent agent process. At every turn, the agent may call only the public laboratory tools. The host validates schemas and preconditions, applies resource accounting, commits successful transactions, and returns public observations. It does not choose or repair an experiment on the participant's behalf. Entity, parametric, and structural campaigns permit eight, ten, and twelve complete experiments, respectively. Checkpoints occur at experiments 0/2/4/6/8, 0/2/4/7/10, and 0/3/6/9/12.

Every checkpoint records model-reliability judgements, query predictions, uncertainty, public evidence references, a typed executable law, and the intended next experiment. Checkpoint repair is part of the evaluated interface burden. All eventually yielded valid records, but rejected attempts remain in the process log and resource accounting.

A.3 Evaluator-owned prediction and law assays

Entity checkpoints contain four prespecified counterfactual queries; parametric and structural checkpoints contain sixteen. The evaluator executes each unique task–world query independently, never returns its truth to the participant, and shares truth across prior arms and checkpoints. The primary readout is mean normalized absolute error across registered query–metric pairs. The executed cohort contains 420/420 truth executions, 1,620 query–metric truth values, 675/675 checkpoints, 6,300 predictions, and 24,300 scored values.

Final typed laws are executed on the same registered coordinates. Schema validity, coordinate coverage, truth-normalized error, change from pre-evidence and effective-final checkpoints, and consistency with explicit final predictions are retained separately. This assay establishes executability and in-domain compression only. Reusability would require new coordinates or transfer to a context-reset process.

A.4 Typed-law capacity and distillation controls

To distinguish participant compression failure from an underpowered output schema, we refitted each of the 135 complete final prediction vectors with legal identity-link laws over the registered feature coordinates and allowed basis functions. A full-basis fit used up to 64 terms per metric. A term-matched fit used exactly the participant's submitted term budget for each metric. A leave-one-query-out fit withheld each registered query in turn and predicted it from all remaining queries. Every fitted law was parsed and executed by the production law evaluator; independently computed design-matrix predictions had to agree with executor output within 10^{-10}.

The participant laws differed from their own final predictions by mean MAE 0.1539. Full-basis fits reproduced 135/135 final prediction states with mean MAE 4.25×10^{-13}; term-matched fits reduced the error to 0.0114 and leave-one-query-out fits reached 0.0788. The control uses the same registered coordinate domain and therefore tests representation capacity and distillation, not global mechanistic identification or transfer.

A.5 Failure-aware inference

The unit of prospective inference is the independently selected task–world cluster, not a checkpoint, query, experiment, or replay. The three prior arms within a cluster are matched. Locus-specific selective-correction contrasts are computed first and are not pooled across different intervention semantics. Failed scientific cells remain assigned. A right-censored session carries its last valid checkpoint forward, and a missing final prediction receives zero primary improvement. Infrastructure recovery is allowed only when no scientific trajectory has persisted and only under the fixed attempt limit.

The matched-evidence structural contrast has five independent worlds. Its interval and exact one-sided sign-flip result are descriptive because the block was designed to localize a transition, not to establish a population-level arm effect. Recipe divergence is likewise a manipulation check: without repeated same-arm sessions, provider stochasticity cannot be separated from intervention sensitivity at exact-recipe resolution.

A.6 Complete-plan action assay

In each open-action cell, a persistent agent performs twelve autonomous experiments and commits checkpoints after 0, 3, 6, 9, and 12 experiments. The host then reveals eight unseen ActionPlans. Each plan specifies ordered operations, all submitted parameters, initial-state assumptions, intermediate measurement positions, terminal assay, and declared omitted operations. Candidate outcomes, ranks, and other-arm evidence stay hidden. The evaluator verifies identity of disclosed, truth-evaluated, and replayed plans.

The primary endpoint is normalized regret of the selected plan within its task–world candidate set. Selected rank, Top-1, pairwise ordering, and law adequacy are secondary or diagnostic. Forty-five cells were scheduled across three tasks, five worlds, and three initial-model arms. Forty-two were eligible. The two agent-induced resource/process failures and one provider interruption are retained as failures in the scheduled denominator and are not replaced. An independent repair is reported only as sensitivity evidence because it follows a new trajectory.

The continuous law–action analysis used only those same 42 originally eligible cells; the three failures stayed in the 45-cell scheduled denominator and were not imputed. Pearson and Spearman associations were computed pooled and by task. Uncertainty resampled the 15 frozen task–world clusters, retaining all eligible arms within a sampled cluster, with 10,000 bootstrap replicates and seed 20260827. The pooled Spearman coefficients were -0.073 for law MAE versus selected rank and -0.133 for law MAE versus normalized regret, with cluster-bootstrap 95% intervals $[-0.380, 0.256]$ and $[-0.452, 0.217]$. A prespecified sensitivity table swept law-MAE thresholds 0.05, 0.075, 0.10, 0.15, 0.20, 0.25, and 0.30 without selecting a cutoff from action outcomes.

A.7 Oracle-control qualification

The unexecuted causal follow-up crossed three tasks, five worlds, three priors, and five conditions: no evidence, stepwise yoked evidence, autonomous exploration, learned law in a fresh context, and a provider-free oracle law. The planned denominator was 225 participant sessions, 45 autonomous donors, and 540 donor experiments. Participant execution required every fresh task–world oracle to pass candidate opportunity checks and to attain Spearman rank correlation at least 0.80 over the eight candidate plans using fit data disjoint from candidate outcomes.

The 96-query qualification stopped after eight of fifteen clusters. It completed 896/896 truth executions and exact replays, with eight candidate gates and seven oracle rank gates passing. The retained crystallization failure had $\rho = 0.738095$, Top-1 disagreement, and zero fit/candidate overlap. No operational canary or participant session began.

The 320-query construction held the fitted ExtraTrees family fixed while expanding coverage to 64 global and 256 candidate-neighborhood queries. A platform-defective partial caused by an evidence-reference schema limit was retained and not reused. After the contract repair, a new root completed

2,352/2,352 truth executions and replays across seven exposed units, passing all seven and repairing four historical failures. The first prospective world then completed 336/336 executions and replays but failed at $\rho = 0.714286$; the remaining fourteen were not started. Candidate outcomes were never read during fitting.

The gate-alignment analysis added no execution. It re-read sixteen frozen unit versions and exactly reproduced their original Spearman and Top-1 values. Its action endpoints were Top-1, normalized regret, selection within 0.01 of optimum, and near-tie-aware pair ordering. It did not alter any historical result, threshold, or stop decision.

A.8 Additional scope boundaries

The study evaluates simulated chemistry and one complete agent–tool system. It does not establish physical laboratory fidelity, universal chemical coverage, cross-provider ordering, or artifact portability. The public cohort is not a private-world confirmation. The 45-cell action matrix lacks a same-agent no-evidence baseline. The five-condition causal cohort is unexecuted. A later reduced development pilot completed no-evidence and learned-law sessions in one electrochemical stratum, but its yoked session was right-censored after five of six provider turns; it is excluded from all main causal and arm-level conclusions.